

# 1. Autonomous Vehicle Vision System Selection

## Comparison:

| Method        | Accuracy | Latency | Computation | Accuracy $\geq 90\%$ ? | Latency $\leq 70$ ms? | Resource Match? | Satisfies All? |
|---------------|----------|---------|-------------|------------------------|-----------------------|-----------------|----------------|
| Viola-Jones   | 82%      | 30 ms   | Low         | No (Fails)             | Yes                   | Yes             | No             |
| YOLO          | 93%      | 60 ms   | Moderate    | Yes                    | Yes                   | Yes             | Yes            |
| Deep Learning | 97%      | 90 ms   | High        | Yes                    | No (Unacceptable)     | No              | No             |



## Logical Justification

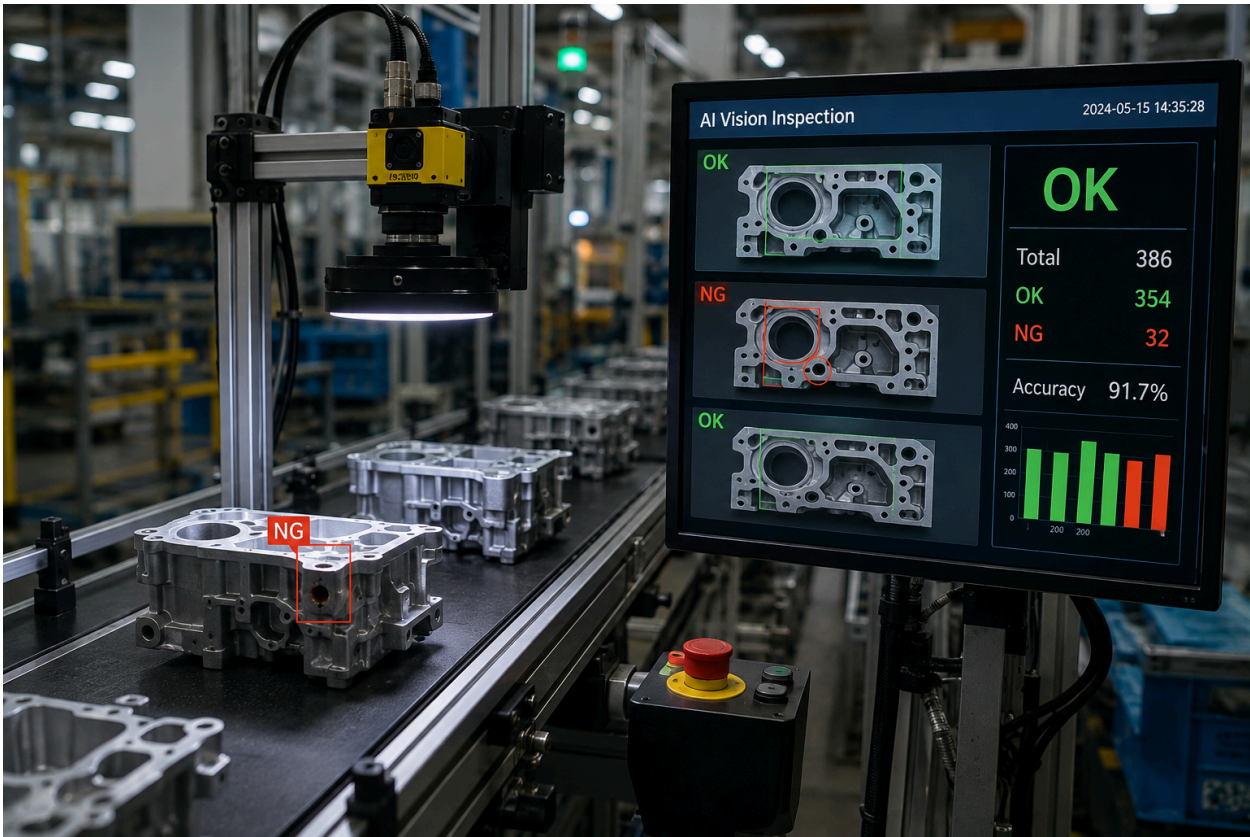
- **Viola-Jones:** Fails immediately on accuracy. At 82%, it falls below the mandatory 90% threshold despite having low latency (30 ms) and light resource usage.
- **Deep Learning:** Fails on speed and hardware capacity. While achieving high accuracy (97%), its 90 ms latency exceeds the maximum 70 ms limit, and its high compute requirements exceed the vehicle's moderate limits.

- **YOLO:** Satisfies all requirements. Its 93% accuracy exceeds the 90% minimum, its 60 ms latency is within the 70 ms limit, and its moderate computational footprint fits the vehicle's hardware constraints.

2. Smart Factory Product Inspection Selection

YOLO is the only method that meets all required constraints for the smart factory product inspection system.

| Method                     | Accuracy | Processing Time | Computation | Accuracy ≥95%?      | Time ≤50 ms?   | Resource Match? | Satisfies All? |
|----------------------------|----------|-----------------|-------------|---------------------|----------------|-----------------|----------------|
| Traditional Image Analysis | 91%      | 25 ms           | Low         | No (Defects Missed) | Yes            | Yes             | No             |
| YOLO                       | 96%      | 45 ms           | Moderate    | Yes                 | Yes            | Yes             | Yes            |
| Deep Learning              | 98%      | 75 ms           | High        | Yes                 | No (Production | No              | No             |



## Logical Justification

- **Traditional Image Analysis:** Fails on accuracy. At 91%, it falls short of the required 95% threshold, increasing the risk of missing defective products despite meeting time (25 ms) and computation limits.
- **Deep Learning:** Fails on speed and hardware requirements. Its 75 ms processing time exceeds the maximum 50 ms limit, risking production bottlenecks, and its high compute requirements exceed the facility's moderate compute support.
- **YOLO:** Satisfies all requirements. Its 96% accuracy exceeds the 95% minimum, its 45 ms processing time stays within the 50 ms budget, and its moderate computational footprint aligns directly with the supported hardware.